

Notes: Malmendier & Tate (JF, 2005)

CEO Overconfidence and Corporate Investment

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1 Big picture

- **Question.** Can CEO overconfidence explain why corporate investment is sensitive to internal cash flow?
- **Core idea.** Overconfident CEOs overestimate the returns to their firms' investment projects and believe the market undervalues their firm. They therefore view external equity as too expensive.
- **Main prediction.** Overconfident CEOs invest more when internal funds are abundant, but they cut back when they must rely on external finance. Their investment is therefore more sensitive to cash flow.
- **Data.** The paper studies CEOs of 477 large publicly traded U.S. firms from 1980 to 1994, using detailed data on CEO stock and option holdings, exercises, firm investment, Q , and cash flow.
- **Overconfidence measurement.** A CEO is classified as overconfident if he persistently fails to reduce firm-specific risk in his personal portfolio. The main measures are *Holder 67*, *Longholder*, and *Net Buyer*.
- **Main result.** Investment-cash flow sensitivity is higher for overconfident CEOs than for non-overconfident CEOs.
- **Financing-constraint result.** The effect is strongest in equity-dependent firms, where perceived equity mispricing is most costly.
- **Alternative-explanation tests.** The paper argues that the option-holding measures are not simply capturing private information, signaling, tax timing, or compensation incentives.
- **Personal traits result.** Education, career background, birth cohort, and CEO role concentration also help explain investment-cash flow sensitivity, but overconfidence remains important after these controls.
- **Policy implication.** Incentive alignment and disclosure may not fully solve investment distortions if managers sincerely but incorrectly believe they are maximizing value.

2 Data and measurement

2.1 Sample construction

- The sample contains large U.S. firms that appear at least four times on Forbes lists of the largest U.S. companies between 1984 and 1994.
- The CEO portfolio data come from Hall and Liebman and Yermack and include option packages, exercise prices, remaining durations, and CEO stock ownership.
- The sample period is 1980 to 1994.
- The merged investment sample contains 337 firms and 3,742 firm-year observations.
- The advantage of the dataset is unusually detailed CEO-level information on personal portfolio decisions, not just compensation totals.

Interpretation: The empirical strategy requires observing whether CEOs personally choose to remain exposed to company-specific risk. Standard firm-level investment data alone would not identify CEO overconfidence.

2.2 Investment, cash flow, capital, and Q

The paper measures:

$$Investment_{it} = \frac{Capital\ Expenditures_{it}}{Capital_{i,t-1}}, \quad (1)$$

$$CashFlow_{it} = \frac{EarningsBeforeExtraordinaryItems_{it} + Depreciation_{it}}{Capital_{i,t-1}}. \quad (2)$$

Capital is property, plant, and equipment. Q is the ratio of market value of assets to book value of assets:

$$Q_{it} = \frac{MarketValueAssets_{it}}{BookValueAssets_{it}}. \quad (3)$$

- Investment is capital expenditures normalized by beginning-of-year capital.
- Cash flow is earnings before extraordinary items plus depreciation, also normalized by beginning-of-year capital.
- Q is included as the main proxy for investment opportunities.
- The authors trim cash flow at the 1% level to reduce the effect of extreme outliers.

Interpretation: The central regressions ask whether cash flow explains investment after controlling for Q and CEO incentives. The overconfidence story predicts a larger cash-flow coefficient for overconfident CEOs.

2.3 CEO option and stock data

- The data include the CEO’s stock ownership and option holdings in each year.
- For each option package, the authors observe the exercise price, remaining duration, and number of underlying shares.
- They do not observe the exact price at which options were exercised; they therefore check robustness under maximum, mean, and median fiscal-year exercise prices.
- CEOs are generally underdiversified because they hold firm stock, nontradable options, and firm-specific human capital.

Interpretation: Because a diversified risk-averse CEO should exercise deep in-the-money options and avoid buying more firm stock, persistent failure to diversify is interpreted as a behavioral signal.

2.4 Overconfidence measure 1: Holder 67

Holder 67 is based on the timing of option exercise after the vesting period.

- The authors examine option packages in year 5, the first year in which packages are at least partially exercisable.
- A package is “deep in the money” if it is more than 67% in the money.
- Hall and Murphy’s executive-option framework suggests that a risk-averse, underdiversified CEO should exercise at such high moneyness.
- A CEO enters the Holder 67 sample restriction if he has at least two such option packages during the sample.
- He is classified as overconfident after the first time he fails to exercise such an option, provided that he later repeats the behavior at least once.

Interpretation: Holder 67 targets habitual late exercise, not a one-time delay. It captures the idea that overconfident CEOs expect the stock price to continue rising under their leadership.

2.5 Overconfidence measure 2: Longholder

- Longholder equals one if a CEO ever holds an option until the final year before expiration.
- The typical option has a ten-year life and is fully vested by year 5.
- Most options held until the final year are deep in the money, with a median moneyness above 250%.
- Longholder is assigned for all CEO-years of CEOs who ever display this behavior.

Interpretation: Longholder is a conservative measure. It does not require choosing a specific moneyness threshold, because holding to the last year is hard to reconcile with standard diversification logic.

2.6 Overconfidence measure 3: Net Buyer

- Net Buyer identifies CEOs who are net purchasers of company stock during their first five years in the sample.
- The measure is applied only to CEOs who remain CEO for at least ten years in the sample.
- The investment regressions exclude the first five years used to classify the CEO.

Interpretation: Net Buyer uses disjoint periods for measurement and outcome, helping reduce the concern that stock purchases and investment sensitivity are driven by the same contemporaneous shock.

2.7 Other CEO personal characteristics

The authors collect education and career data from Dun and Bradstreet and Who's Who in Finance and Industry.

- Finance education includes accounting, finance, business, and economics degrees.
- Technical education includes engineering and natural science degrees.
- Finance career includes prior work in financial institutions or finance-related roles.
- Technical career includes engineering, science, or patent-related backgrounds.
- The paper also tracks whether CEOs were born in the 1930s and whether they hold multiple top positions such as CEO, president, and chairman.

Interpretation: These variables test whether overconfidence is merely proxying for broader CEO style or background.

3 Model and empirical specifications

3.1 Model primitives

The firm has existing assets A and s shares outstanding. At time 1 cash flow C is realized. The CEO chooses investment I , cash financing c , debt financing d , and equity issuance s' . Expected project return is $R(I)$ with

$$R'(I) > 0, \quad R''(I) < 0. \quad (4)$$

An overconfident CEO overestimates returns by percentage Δ , so perceived expected return is

$$R(I)(1 + \Delta). \quad (5)$$

Interpretation: The only friction in the model is distorted managerial beliefs. There is no agency problem or asymmetric information in the baseline, which makes the overconfidence mechanism clean.

3.2 CEO maximization problem

The CEO maximizes the perceived value of current shareholders:

$$\max_{I, s', c, d} \frac{s}{s + s'} [A + C + R(I)(1 + \Delta) - c - d]. \quad (6)$$

New shareholders require fair compensation based on true expected value:

$$\frac{s'}{s + s'} [A + C + R(I) - c - d] = I - c - d. \quad (7)$$

Financing constraints are

$$c \leq C, \quad d \leq D, \quad c + d \leq I, \quad c \geq 0, \quad d \geq 0, \quad I \geq 0. \quad (8)$$

Interpretation: The CEO thinks equity is undervalued because he overestimates project returns. Cash and risk-free debt do not create the same disagreement. This is why internal cash flow matters.

3.3 First-best and desired investment

The first-best investment level satisfies

$$R'(I_{FB}) = 1. \quad (9)$$

The overconfident CEO's desired unconstrained investment level satisfies

$$R'(\hat{I}) = \frac{1}{1 + \Delta}. \quad (10)$$

Because $\Delta > 0$, the overconfident CEO desires

$$\hat{I} > I_{FB}. \quad (11)$$

Interpretation: Overconfidence lowers the CEO's perceived marginal cost of investment relative to perceived marginal return, so he wants to invest more than the first-best.

3.4 Lemma 1

Lemma 1.

- If the CEO is rational, $\Delta = 0$, then $I^* = I_{FB}$ for all C and D .
- If the CEO is overconfident, $\Delta > 0$, then $I^* = \hat{I} > I_{FB}$ if $C + D \geq \hat{I}$.
- If the CEO is overconfident and $C + D < \hat{I}$, then $\hat{I} > I^* > I_{FB}$.

Interpretation: With enough internal cash and riskless debt capacity, the overconfident CEO overinvests fully. When constrained, he still overinvests relative to first-best but cannot reach his desired level.

3.5 Proposition 1

Proposition 1.

- If the CEO is rational, investment is independent of internal cash flow.
- If the CEO is overconfident, investment is independent of cash flow when $C + D \geq \hat{I}$.
- If the CEO is overconfident and $C + D < \hat{I}$, investment is strictly increasing in cash flow.

Interpretation: Cash flow matters for overconfident CEOs only when internal cash plus debt capacity is insufficient. This generates the paper’s two main empirical predictions.

3.6 Prediction 1

$$\left. \frac{\partial Investment}{\partial CashFlow} \right|_{Overconfident} > \left. \frac{\partial Investment}{\partial CashFlow} \right|_{Not\ Overconfident}. \quad (12)$$

Interpretation: Overconfident CEOs are more reluctant to issue equity, so their investment depends more on internal funds.

3.7 Prediction 2

$$\text{Overconfidence effect is stronger in equity-dependent firms.} \quad (13)$$

Interpretation: If a firm has sufficient debt capacity, the CEO can finance investment without issuing equity. If the firm is equity-dependent, the perceived undervaluation of equity becomes binding.

3.8 Baseline investment regression

The paper estimates variants of

$$Investment_{it} = \alpha_i + \lambda_t + \beta_1 CashFlow_{it} + \beta_2 Q_{it} + \beta_3 X_{it} \quad (14)$$

$$+ \beta_4 Overconfidence_{it} + \beta_5 (Overconfidence_{it} \times CashFlow_{it}) + \epsilon_{it}. \quad (15)$$

Controls include CEO stock ownership, vested options, firm size, corporate governance, and interactions of these variables with cash flow.

Interpretation: The coefficient of interest is β_5 . The overconfidence hypothesis predicts $\beta_5 > 0$.

3.9 Equity-dependence regressions

For financing constraints, the paper splits firms into Kaplan-Zingales index quintiles and estimates the same investment regression in each quintile.

Interpretation: Prediction 2 implies that the overconfidence-cash flow interaction should be largest in the most constrained quintile.

4 Identification and empirical design

4.1 What identifies the main effect

- The key variation is CEO-level overconfidence measured from personal portfolio decisions.
- The outcome is firm investment sensitivity to cash flow.
- The regressions include Q to control for investment opportunities.
- Firm and year fixed effects absorb permanent firm heterogeneity and aggregate investment conditions.
- CEO controls absorb incentives and governance channels.

Interpretation: The design asks whether two CEOs facing similar firm characteristics differ in investment-cash flow sensitivity because one persistently fails to diversify.

4.2 Why option-holding behavior is informative

- CEOs are underdiversified due to firm stock, nontradable options, and firm-specific human capital.
- A rational risk-averse CEO should exercise deep in-the-money options early.
- Persistent late exercise suggests excessive optimism about future firm performance.
- Persistent buying of company stock suggests the same belief.

Interpretation: The measures are behavioral: they infer beliefs from costly personal portfolio choices.

4.3 Alternative explanation: inside information

- If CEOs hold options because of positive private information, they should earn abnormal returns from holding rather than exercising.
- The paper compares actual option-holding returns to the counterfactual of exercising and investing in the S&P 500.
- CEOs who hold options beyond thresholds do not, on average, beat the market.
- Late exercise is persistent across years, suggesting a personal fixed effect rather than time-varying information.

Interpretation: The information story has difficulty explaining persistent late exercise without abnormal profits.

4.4 Alternative explanation: signaling

- CEOs may hold options to signal confidence in the firm.
- However, if signaling reduces asymmetric information, it should reduce investment-cash flow sensitivity, not increase it.
- Option exercises are also not widely interpreted as strong market signals compared with open-market stock trades.

Interpretation: The investment regression itself helps reject the pure signaling interpretation.

4.5 Alternative explanation: incentive alignment

- Holding more stock or options could align CEO incentives with shareholders.
- The regressions control for stock ownership and vested option holdings, including interactions with cash flow.
- Overconfidence remains important after controlling for these incentive variables.

Interpretation: The results are not just that CEOs with stronger incentives invest differently; they are tied to the behavioral overconfidence measures.

4.6 Endogeneity and limits

- Boards may select CEOs partly because of personality traits, including confidence.
- Overconfidence may correlate with unobserved firm strategy or industry dynamics.

- The authors use industry controls, firm fixed effects where possible, year fixed effects, and CEO characteristics to reduce these concerns.
- Even if boards choose overconfident CEOs intentionally, the paper’s central point remains: CEO beliefs shape investment policy.

Interpretation: The evidence is strongest as proof that CEO behavioral traits matter for investment, not as a randomized assignment of CEO overconfidence.

5 Tables and figure

5.1 Table I: Summary statistics

Read:

- The full investment sample contains 337 firms and 3,742 firm-year observations.
- The Holder 67 sample restriction contains 113 CEOs and 1,058 CEO-years.
- The Holder 67 overconfident sample contains 58 CEOs and 305 CEO-years.
- Firm variables such as assets, capital, cash flow, Q, ROA, and industry distribution look broadly similar across the full sample and the Holder subsamples.
- CEO age, tenure, stock ownership, vested options, and governance variables are also summarized for the full and Holder samples.

Economic message: The table supports the claim that measured overconfidence is not obviously just a proxy for observable firm characteristics or industry composition.

Table I
Summary Statistics

The sample with "Holder 67 Sample Restriction" contains all CEO-years of CEOs who had options more than 67% in the money in the fifth year at least two times during their sample tenure. The "Holder 67 Sample" contains all CEO-years after the CEO fails to exercise a five-year-old option that is at least 67% in-the-money, provided that he subsequently does it again at least once.

Variable	Panel A: Firm Data Summary Statistics											
	Full Sample Number of Firms = 337				Holder 67 Sample Restriction Number of Firms = 113				Holder 67 Sample Number of Firms = 58			
	Obs.	Mean	Median	SD	Min.	Max.	Obs.	Mean	Median	SD	Min.	Max.
Assets (\$M)	3742	5,603	2,296	12,700	14	189,509	3058	5,568	2,161	12,978	20	180,237
Capital (\$M)	3740	2,448	889	5,599	4	128,963	3058	1,831	846	3,616	4	43,027
Investment (\$M)	3742	392	151	949	0	17,630	3058	270	159	878	0	11,712
Investment normalized by lagged capital	3742	0.23	0.18	0.25	0	5.72	3058	0.25	0.21	0.23	0	5.72
Investment normalized by lagged assets	3742	0.09	0.07	0.08	0	1.64	3058	0.10	0.08	0.07	0	1.64
Cash flow (\$M)	3742	453	191	885	-618	15,738	3058	490	203	1,005	-117	11,713
Cash flow normalized by lagged capital	3742	0.35	0.25	0.35	-0.24	2.55	3058	0.42	0.32	0.35	-0.11	2.46
Cash flow normalized by lagged assets	3742	0.11	0.10	0.07	-0.16	0.65	3058	0.13	0.12	0.07	-0.06	0.49
Q beginning of the fiscal year	3742	1.44	1.14	0.92	0.51	12.39	3058	1.58	1.31	0.90	0.68	10.71
Return/Price Ratio	3648	0.05	0.06	0.15	-7	1	3058	0.08	0.06	0.06	-0.84	0.85
ROA	3742	0.06	0.05	0.06	-0.21	0.55	3058	0.08	0.07	0.07	-0.19	0.55
Corporate governance (outside CEOs)	3742	1.77	1	1.89	0	9	3058	1.74	1	1.88	0	9

Distribution across Fama-French 12 Industry Groups (3,728 Observations)

Consumer nondurables	0.11	Telecommunication	0.04	Cons. ND	0.14	Telecomm.	0.02	Cons. ND	0.10	Telecomm.	0.02	
Consumer durables	0.04	Utilities	0.17	Cons. D	0.05	Utilities	0.03	Cons. D	0.05	Utilities	0.00	
Manufacturing	0.13	Shops	0.11	Manuf.	0.12	Shops	0.12	Manuf.	0.10	Shops	0.12	
Energy	0.03	Health	0.05	Energy	0.01	Health	0.09	Energy	0.02	Health	0.10	
Chemicals and allied products	0.06	Money	0.06	Chemicals	0.12	Money	0.05	Chemicals	0.15	Money	0.05	
Business equipment	0.07	Other	0.26	Bus. Equip.	0.12	Other	0.26	Bus. Equip.	0.16	Other	0.25	

The Fama-French industries are defined on French's website, (http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html).

Table I, Panel A: firm summary statistics

Table I—Continued

Panel B: CEO Data Summary Statistics

Variables	Panel B: CEO Data Summary Statistics											
	Full Sample Number of CEOs = 497				Holder 67 Sample Restriction Number of CEOs = 113				Holder 67 Sample Number of CEOs = 58			
	Obs.	Mean	Median	SD	Min.	Max.	Obs.	Mean	Median	SD	Min.	Max.
Age	3741	57.57	58	6.77	23	84	3058	56.02	58	6.12	41	82
Years as CEO	3736	8.50	6	7.44	1	45	3058	10.81	9	7.24	1	38
CEO & President & Chairman	3742	0.38	0	0.49	0	1	3058	0.37	0	0.48	0	1
Founder (dummy)	3186	0.17	0	0.37	0	1	3058	0.17	0	0.38	0	1
Stock ownership (%)	3742	0.023	0.0002	0.07	0	0.951	3058	0.017	0.0002	0.05	0	0.38
Vested options (% of shares outst.)	3742	0.003	0.0005	0.01	0	0.483	3058	0.004	0.001	0.01	0	0.11
"Depression baby" (born in 1930s) (dummy)	3741	0.27	0	0.44	0	1	3058	0.42	0	0.49	0	1
Finance career (dummy)	2014	0.23	0	0.42	0	1	493	0.21	0	0.41	0	1
Technical career (dummy)	2014	0.19	0	0.39	0	1	493	0.18	0	0.38	0	1
Finance education (dummy)	2218	0.23	0	0.42	0	1	798	0.38	0	0.49	0	1

Table I, Panel B: CEO summary statistics

5.2 Table II: Longholder and Net Buyer

Read:

- Table II describes the alternative overconfidence measures Longholder and Net Buyer.
- Longholder is conservative: 85 of 661 CEOs are classified as overconfident under this measure.
- Net Buyer is more restrictive in sample size because it requires long CEO tenure and disjoint classification/outcome periods.
- The table also reports correlations between measures, showing that the overconfidence proxies are related but not identical.

Economic message: The paper does not rely on one narrow behavioral measure. Similar investment results across multiple measures make the overconfidence interpretation more credible.

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Table III
Persistence of Exercising Behavior

In Panel A, the dependent variable is a dummy variable taking the value one if the CEO fails to exercise a five-year-old option that reaches at least 67% in-the-money in the current period. Past late exercises are the number of times that the CEO has exercised such options late in the past. Q is the market value of assets over the book value of assets at the beginning of the year. Earnings/Price ratio is the minimal earnings to price ratio during the fiscal year. Panel B presents statistics on late exercises of stock options partitioned by the number of past late exercises by the CEO in question.

Panel A. Random Effects Probit Regression				
Sample: Observations with 67% in-the-Money Options (in Year Five)				
	(1)	(2)	(3)	(4)
Past late exercises	0.2493 (4.40)***	0.2569 (4.57)***	0.2571 (4.61)***	0.266 (4.80)***
Q		-0.1519 (1.79) [*]		-0.1514 (1.81) [*]
Earnings/price ratio			-0.709 (0.77)	-0.8128 (0.89)
Observations	759	742	731	728
Number of CEOs	278	273	272	271

Panel B. Percent of "Late Exercisers" Partitioned by Number of Past Late Exercises

Sample: Observations with 67% in-the-Money Options (in Year Five)		
Past Late Exercises	% Who Exercise Late	Number of CEOs
0	0.32	487
1	0.64	128
2	0.73	67
3	0.94	32
4	0.79	28
>4	0.74	23

Absolute value of z statistics in parentheses.

* Significant at 10%; ** significant at 5%; *** significant at 1%.

such an option in the past.¹⁴ The sample consists of the 759 firm-years in which a CEO had options beyond the 67% threshold in the fifth year after the grant date. The dependent variable is equal to one whenever the CEO did not exercise any portion of those options prior to that year. Panel A of Table III presents the regression results. In column 1, the coefficient of "past late exercises" is positive (0.2493) and highly significant ($z = 4.40$). This implies that the typical CEO, rather than varying his exercise behavior over time, either persistently holds options beyond the 67% threshold or persistently exercises early. The results are robust to the inclusion of Q (column 2) and the firm's earnings–price ratio (column 3) as controls. Consistent with Jenter (2002), high values of Q —perhaps, in this context, a proxy for market overvaluation—appear to decrease

¹⁴ The results are the same if we look only at CEOs who had an option to reach the 67% threshold at least twice.

Table II
Correlations

Sample A: CEOs with options more than 67% in-the-money in the fifth year at least two times and with at least 10 years in the data and no more than one missing observation of ownership in the first five years. Data from the first five years of each CEO's tenure excluded.

Sample B: CEOs with at least 10 years in the data and no more than one missing observation of ownership in the first five years. Data from the first five years of each CEO's tenure excluded.

Sample C: CEOs with options more than 67% in-the-money in the fifth year at least two times and with at least 10 years in the data and no more than one missing observation of ownership in the first five years. Data from the first five years of each CEO's tenure excluded.

	Holder 67			Longholder			Not Holder			Size			Q			CFR			Stock Ownership			Vested Options			Corp. Chg.		
	A	B	C	A	B	C	A	B	C	A	B	C	A	B	C	A	B	C	A	B	C	A	B	C	A	B	C
Holder 67	1.00			1.00			1.00			1.00			1.00			1.00			1.00			1.00			1.00		
Longholder	0.25	1.00		0.25	1.00		0.25	1.00		0.25	1.00		0.25	1.00		0.25	1.00		0.25	1.00		0.25	1.00		0.25	1.00	
Not Holder	0.05	0.05	1.00	0.05	0.05	1.00	0.05	0.05	1.00	0.05	0.05	1.00	0.05	0.05	1.00	0.05	0.05	1.00	0.05	0.05	1.00	0.05	0.05	1.00	0.05	0.05	1.00
Size	0.22	0.25	0.06	-0.17	-0.17	0.06	0.22	0.25	0.06	0.22	0.25	0.06	0.22	0.25	0.06	0.22	0.25	0.06	0.22	0.25	0.06	0.22	0.25	0.06	0.22	0.25	0.06
Q	-0.03	0.05	-0.04	0.10	0.10	-0.04	-0.11	-0.11	-0.04	-0.11	-0.11	-0.04	-0.11	-0.11	-0.04	-0.11	-0.11	-0.04	-0.11	-0.11	-0.04	-0.11	-0.11	-0.04	-0.11	-0.11	-0.04
CFR	-0.03	0.05	-0.04	0.10	0.10	-0.04	-0.11	-0.11	-0.04	-0.11	-0.11	-0.04	-0.11	-0.11	-0.04	-0.11	-0.11	-0.04	-0.11	-0.11	-0.04	-0.11	-0.11	-0.04	-0.11	-0.11	-0.04
Stock	-0.17	0.05	-0.17	-0.17	0.05	-0.17	-0.17	0.05	-0.17	-0.17	0.05	-0.17	-0.17	0.05	-0.17	-0.17	0.05	-0.17	-0.17	0.05	-0.17	-0.17	0.05	-0.17	-0.17	0.05	-0.17
Vested options	0.05	0.05	0.17	0.05	0.05	0.17	0.05	0.05	0.17	0.05	0.05	0.17	0.05	0.05	0.17	0.05	0.05	0.17	0.05	0.05	0.17	0.05	0.05	0.17	0.05	0.05	0.17
Corporate governance	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05

Coefficients in italics are significant at 10%.

5.3 Table III: Persistence of exercising behavior

Read:

- Panel A estimates a random effects probit for whether a CEO holds a five-year-old option that is at least 67% in the money.
- The number of past late exercises strongly predicts current late exercise.
- The coefficient on past late exercises is about 0.25 and highly significant.
- Q has a weak negative effect, and the earnings-price ratio is not statistically important.
- Panel B shows that the probability of late exercise rises sharply with the number of past late exercises.

Economic message: Late exercise looks like a persistent CEO trait rather than a sequence of independent information events.

5.4 Table IV: Returns from holding options too long

Read:

- Table IV compares actual returns from holding deep in-the-money options with counterfactual returns from exercising and investing in the S&P 500.

- Across maximum, mean, and median exercise-price assumptions, CEOs do not earn reliably positive abnormal returns from holding.
- The average CEO beats the S&P 500 only about half the time.
- This weakens the interpretation that late exercise reflects good private information.

Economic message: If late exercise were mainly private information, CEOs should profit from it. The table shows little evidence that they do.

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Table IV
Distribution of Returns of "Late Exercisers" (67%, Fifth Year)

The table presents data on the returns of late exercising CEOs (Holders 67) by percentiles. The first column presents the percentage in-the-money at the maximum price during the fifth fiscal year from grant date for each option package that is held beyond the 67% threshold. The second, third, and fourth columns present the returns (in %) relative to exercising the options during year 5 and investing instead in the S&P 500, assuming exercise at the maximum, mean, and median stock prices during the fiscal year, respectively. We also present the last percentile for which the return is negative under each price assumption. All returns are annualized. **Sample:** CEOs who have option packages at least 67% in-the-money in the fifth year after the option grant and who have not exercised the options before the fifth year.

Percentage in-the-Money in Year 5		Return (in %) Relative to Exercising during Year 5 and Investing in S&P 500					
		Exercise at Fiscal-Year Maximum Price		Exercise at Fiscal-Year Mean Price		Exercise at Fiscal-Year Median Price	
		Percentile	Return	Percentile	Return	Percentile	Return
10th	161.89	10th	-16.56	10th	-16.48	10th	-16.45
20th	213.71	20th	-10.32	20th	-10.51	20th	-11.65
30th	280.97	30th	-6.40	30th	-5.89	30th	-7.39
40th	366.88	40th	-2.79	40th	-2.50	40th	-2.56
		46th	-0.66	46th	-0.38	49th	-0.05
50th	435.88	50th	1.02	50th	1.64	50th	0.30
60th	616.83	60th	5.72	60th	6.94	60th	5.59
70th	905.43	70th	10.86	70th	10.96	70th	11.62
80th	1,395.22	80th	19.16	80th	17.32	80th	16.05
90th	2,326.39	90th	28.27	90th	25.27	90th	25.07
Mean	1,275.90		3.60		4.85		3.57
Standard deviation	3,336.66		20.23		20.96		21.15
Observations	182		182		182		182
CEOs	86		86		86		86

future stock prices. They point to vesting and expiration times as the main determinants of exercise.¹⁵ Our only stock-based proxy for overconfidence, *Net Buyer*, measures overconfidence and investment-cash flow sensitivity for two disjoint time periods to specifically eliminate the influence of signaling.

The same arguments address the hypothesis that CEOs use their insider trades opportunistically to raise the capital market valuation of their firms.

3. *Risk tolerance.* Alternatively, one might want to interpret our measures of overconfidence as measures of risk attitude. A CEO may hold his options beyond

¹⁵ Financial advisory firms that track insider trades often report only the purchases and sales of stocks, not the exercise of options (see, for instance, the websites of Investor and Quicken; see also EDGAR). Other websites make it explicit that option exercise should be understood as "noise" with respect to insider knowledge; see, for instance, <http://www.winninginvesting.com/insider.htm> ("The employees consider the options part of their salary...") or <http://invest-faq.com/articles/trade-insider.html> ("same-day exercise of a stock option and selling the resulting stock ... rarely means very much."); similarly, see the weekly column "Insider Trading Spotlight" in the *Wall Street Journal*, and the weekly coverage of insider-trading information in the *Financial Times*.

5.5 Table V and Figure 1: Main investment-cash flow regressions

Read:

- Table V estimates investment regressions using Holder 67, Longholder, and Net Buyer measures.
- Cash flow has a positive and significant baseline effect on investment.
- The interaction between Holder 67 and cash flow is positive and significant in the main specifications.
- The Longholder-cash flow interaction is also positive and significant in the fixed-effects specifications.
- The Net Buyer-cash flow interaction is positive and significant in several specifications, especially when allowing industry-cash flow interactions.
- Figure 1 shows that the Holder threshold result is robust across many moneyiness cutoffs, not just 67%.

Economic message: This is the central empirical result: overconfident CEOs have higher investment-cash flow sensitivity.

Table V
Regression of Investment on Cash Flow and Exercise Behavior
The dependent variable in the regressions is Investment, defined as firm capital expenditures and normalized by capital at the beginning of the year. Cash flow is earnings before extraordinary items plus depreciation and is normalized by capital at the beginning of the year. Q is the market value of assets over the book value of assets at the beginning of the year. Stock ownership is the fraction of company stock owned by the CEO and his immediate family at the beginning of the year. Vested options are the CEO's holdings of options that are exercisable within 6 months of the beginning of the year, as a fraction of common shares outstanding. Vested options are multiplied by 10 so that the mean is comparable to stock ownership. Size is the natural logarithm of assets at the beginning of the year. Corporate governance is the number of outside directors who currently serve as CEOs of other companies. *Holder 67* is a dummy variable equal to one for all CEO-years after the CEO holds a five-year-old option that is more than 67% in-the-money, provided that he subsequently does it again at least once. Industries are defined as the 12 Fama-French industry groups. In columns 6 and 7, standard errors are robust to heteroskedasticity and arbitrary within-firm serial correlation.
Sample: CEOs with options more than 67% in-the-money in the fifth year at least two times.

	Late Exercise of 67%-in-the-Money Options (in Year 5)					
	Baseline Regressions	Overconfidence with Fixed Effects, No Controls (1)	Overconfidence with Fixed Effects, No Controls (2)	Overconfidence with Fixed Effects, No Controls (3)	Standard Errors Clustered by Firm (6)	Industry-CF Interactions, Clustered by Firm (7)
Cash flow	0.2052 (9.73)***	0.6419 (7.59)***	1.6579 (9.85)***	0.6729 (7.56)***	1.7044 (10.26)***	1.2911 (3.22)***
Q	0.0250 (3.04)***	0.0635 (6.54)***	0.0656 (0.34)	-0.0049 (6.79)***	-0.0088 (0.44)	-0.0112 (0.18)
Stock ownership (%)		-0.1077 (0.19)	-0.1077 (0.19)	-0.1077 (0.19)	-0.1854 (0.35)	0.1892 (0.34)
Vested options		0.1946 (1.61)	0.1946 (1.61)	0.1946 (1.61)	0.1398 (1.17)	0.1589 (1.55)
Size		0.0466 (2.45)**	0.0466 (2.45)**	0.0543 (2.86)***	0.0543 (1.47)	0.0429 (1.44)
Corporate governance		-0.0042 (0.54)	-0.0042 (0.54)	-0.0071 (0.92)	-0.0071 (0.70)	-0.0131 (1.40)

(continued)

CEO Overconfidence and Corporate Investment

Table V—Continued

	Late Exercise of 67%-in-the-Money Options (in Year 5)					
	Baseline Regressions	Overconfidence with Fixed Effects, No Controls (1)	Overconfidence with Fixed Effects, No Controls (2)	Overconfidence with Fixed Effects, No Controls (3)	Standard Errors Clustered by Firm (6)	Industry-CF Interactions, Clustered by Firm (7)
Q + (Cash flow)		0.0521 (2.64)***	0.0521 (2.64)***	0.0521 (2.64)***	0.0648 (0.83)	0.0845 (1.28)
(Stock ownership) + (Cash flow)		-0.3749 (1.38)	-0.3749 (1.38)	-0.3749 (1.38)	-0.6897 (0.45)	-1.1158 (0.97)
(Vested options) + (Cash flow)		-0.4612 (4.15)***	-0.4612 (4.15)***	-0.4612 (4.15)***	-0.2981 (1.32)	-0.5015 (2.45)***
(Size) + (Cash flow)		-0.1713 (8.47)***	-0.1713 (8.47)***	-0.1713 (8.47)***	-0.1754 (0.94)	-0.1433 (0.89)
(Corporate governance) + (Cash flow)		0.0383 (2.16)**	0.0383 (2.16)**	0.0383 (2.16)**	0.0441 (0.65)	0.0597 (0.61)
<i>Holder 67</i>		-0.0351 (1.35)	-0.0351 (1.35)	-0.0351 (1.35)	-0.0405 (1.96)*	-0.0362 (1.27)
(<i>Holder 67</i>) + (Cash flow)		0.1648 (3.89)***	0.1648 (3.89)***	0.1648 (3.89)***	0.2339 (4.70)***	0.1718 (2.39)**
Year-fixed effects	no	yes	yes	yes	yes	yes
Firm-fixed effects	no	yes	yes	yes	yes	yes
(Year-fixed effects) + (Cash flow)	no	yes	yes	yes	yes	yes
(Year-fixed effects) + (Cash flow)	no	yes	yes	yes	yes	yes
Observations	1058	1058	1058	1058	1058	1058
Adjusted R^2	0.13	0.56	0.61	0.56	0.62	0.67

Table V, first page: main regressions

Table V, continued

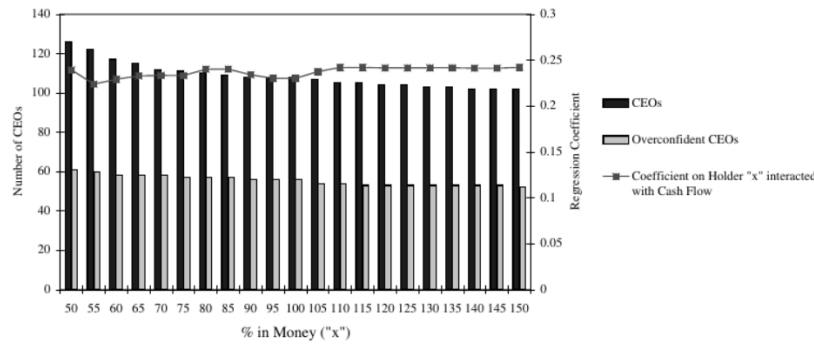


Figure 1. Holder regression for different % in-the-money. Figure 1 presents the results of reestimating the regression specified in column 6 of Table V using different percentages in-the-money as thresholds for rational exercise in the classification of CEOs as overconfident. More specifically, *Holder 67* is replaced in the regression by *Holder "x"*, where *Holder "x"* is a dummy variable equal to one for all CEO-years after the CEO holds a five-year-old option that is more than "x%" in-the-money, provided that he subsequently does it again at least once. In addition, the sample is restricted in each regression to the subsample of CEOs who at least twice had options that reached at least "x%" in the money after 5 years. The number of CEOs meeting this restriction for each "x" is presented below along with the subset of those CEOs who are classified as overconfident using the *Holder "x"* measure. Coefficients on *Holder "x"* interacted with cash flow are significant at the 5% level for all x except x = 50, 80, and 85, which are significant at 1%, where standard errors are robust to heteroskedasticity and arbitrary within-firm serial correlation.

5.6 Table VI: Separating overconfidence from information

Read:

- Table VI splits late exercisers into CEOs who won by holding and CEOs who lost by holding.
- If private information drove the result, investment-cash flow sensitivity should be strongest among winners.
- The Hold-and-Lose 67 interaction with cash flow remains positive and significant.
- Thus, even CEOs who lost money by holding options still display the overconfidence investment pattern.

Economic message: The table directly attacks the inside-information explanation. The effect is present among CEOs whose option-holding choice turned out poorly.

Table VI
Regression of Investment on Cash Flow and Exercise Behavior

The dependent variable in the regressions is investment, defined as firm capital expenditures and normalized by capital at the beginning of the fiscal year. Cash flow is earnings before extraordinary items plus depreciation and is normalized by capital at the beginning of the fiscal year. Q is the market value of assets over the book value of assets at the beginning of the year. Stock ownership is the fraction of company stock owned by the CEO and his immediate family at the beginning of the year. Vested options are the CEO's holdings of options that are exercisable within 6 months of the beginning of the year, as a fraction of common shares outstanding. Vested options are multiplied by 10 so that the mean is comparable to stock ownership. Size is the natural logarithm of assets at the beginning of the fiscal year. Corporate governance is the number of outside directors who currently serve as CEOs of other companies. *Hold and Win 67* is a dummy variable equal to one for all CEO years after the CEO holds a five-year-old option that is more than 67% in-the-money, provided that he subsequently does it again at least once and that he earns excess returns by holding the options (relative to exercising in the fifth year and investing the proceeds in the S&P 500) each time. *Hold and Low 67* is a dummy variable equal to one for all CEO years after the CEO holds a five-year-old option that is more than 67% in-the-money, provided that he subsequently does it again at least once and that he loses money by holding such an option (relative to exercising in the fifth year and investing the proceeds in the S&P 500) at least once. Returns are calculated using the maximum stock price during the fiscal year. Industries are defined as the 12 Fama-French industry groups. In columns 6 and 7, standard errors are robust to heteroskedasticity and arbitrary within-firm serial correlation.

	Baseline Regressions			Overconfidence with Fixed Effects, No Controls			Overconfidence with Fixed Effects, No Controls			Standard Errors Clustered by Firm			Industry-CF Interactions, Clustered by Firm		
	No Fixed Effects, No Controls (1)	Fixed Effects, No Controls (2)	Fixed Effects, No Controls (3)	Overconfidence with Fixed Effects, No Controls (4)	Overconfidence with Fixed Effects, No Controls (5)	Standard Errors Clustered by Firm (6)	Overconfidence with Fixed Effects, No Controls (7)	Overconfidence with Fixed Effects, No Controls (8)	Standard Errors Clustered by Firm (9)	Industry-CF Interactions, Clustered by Firm (10)	Industry-CF Interactions, Clustered by Firm (11)	Industry-CF Interactions, Clustered by Firm (12)	Industry-CF Interactions, Clustered by Firm (13)	Industry-CF Interactions, Clustered by Firm (14)	Industry-CF Interactions, Clustered by Firm (15)
Cash flow	0.2096 (9.58)***	0.0576 (7.14)***	1.7235 (9.82)***	0.6951 (7.42)***	1.7898 (9.95)***	1.7296 (2.96)***	1.7898 (3.21)***	1.7296 (3.21)***	1.7898 (3.21)***	1.7296 (3.21)***	1.7898 (3.21)***	1.7296 (3.21)***	1.7898 (3.21)***	1.7296 (3.21)***	1.7898 (3.21)***
Q	0.0271 (3.14)***	0.0668 (6.20)***	-0.0057 (0.30)	0.0718 (6.74)***	-0.007 (0.32)	-0.007 (0.30)	-0.007 (0.32)	-0.007 (0.30)	-0.007 (0.32)	-0.007 (0.30)	-0.007 (0.32)	-0.007 (0.30)	-0.007 (0.32)	-0.007 (0.30)	-0.007 (0.32)
Stock ownership (%)			-0.2164 (0.31)		-0.2233 (0.31)	-0.2233 (0.31)		-0.2233 (0.31)	-0.2233 (0.31)		-0.2233 (0.31)	-0.2233 (0.31)		-0.2233 (0.31)	-0.2233 (0.31)
Vested options			0.1802 (1.46)		0.1111 (1.07)	0.1111 (0.96)		0.1111 (1.07)	0.1111 (0.96)		0.1111 (1.07)	0.1111 (0.96)		0.1111 (1.07)	0.1111 (0.96)
Size			0.0535 (2.71)***		0.0587 (2.99)***	0.0587 (2.92)		0.0587 (2.92)	0.0587 (2.92)		0.0587 (2.92)	0.0587 (2.92)		0.0587 (2.92)	0.0587 (2.92)

(continued)

Table VI—Continued

	Baseline Regressions			Overconfidence with Fixed Effects, No Controls			Overconfidence with Fixed Effects, No Controls			Standard Errors Clustered by Firm			Industry-CF Interactions, Clustered by Firm		
	No Fixed Effects, No Controls (1)	Fixed Effects, No Controls (2)	Fixed Effects, No Controls (3)	Overconfidence with Fixed Effects, No Controls (4)	Overconfidence with Fixed Effects, No Controls (5)	Standard Errors Clustered by Firm (6)	Overconfidence with Fixed Effects, No Controls (7)	Overconfidence with Fixed Effects, No Controls (8)	Standard Errors Clustered by Firm (9)	Industry-CF Interactions, Clustered by Firm (10)	Industry-CF Interactions, Clustered by Firm (11)	Industry-CF Interactions, Clustered by Firm (12)	Industry-CF Interactions, Clustered by Firm (13)	Industry-CF Interactions, Clustered by Firm (14)	Industry-CF Interactions, Clustered by Firm (15)
Corporate governance			-0.0042 (0.52)		-0.0072 (0.90)	-0.0072 (0.75)		-0.0072 (0.90)	-0.0072 (0.75)		-0.0072 (0.90)	-0.0072 (0.75)		-0.0072 (0.90)	-0.0072 (0.75)
$Q \times$ (Cash flow)			0.0027 (2.53)***		0.0047 (3.10)***	0.0047 (2.82)		0.0047 (3.10)***	0.0047 (2.82)		0.0047 (3.10)***	0.0047 (2.82)		0.0047 (3.10)***	0.0047 (2.82)
(Stock ownership) $\times$ (Cash flow)			-0.1001 (1.71)*		-0.0033 (1.92)*	-0.0033 (1.55)		-0.0033 (1.92)*	-0.0033 (1.55)		-0.0033 (1.92)*	-0.0033 (1.55)		-0.0033 (1.92)*	-0.0033 (1.55)
(Vested options) $\times$ (Cash flow)			-0.4438 (8.87)***		-0.2907 (3.46)***	-0.2907 (3.20)		-0.2907 (3.46)***	-0.2907 (3.20)		-0.2907 (3.46)***	-0.2907 (3.20)		-0.2907 (3.46)***	-0.2907 (3.20)
(Size) $\times$ (Cash flow)			-0.1761 (8.42)***		-0.1768 (8.46)***	-0.1768 (8.46)		-0.1768 (8.46)***	-0.1768 (8.46)		-0.1768 (8.46)***	-0.1768 (8.46)		-0.1768 (8.46)***	-0.1768 (8.46)
(Corporate governance) $\times$ (Cash flow)			0.0083 (2.10)**		0.0047 (2.07)**	0.0047 (1.72)*		0.0047 (2.07)**	0.0047 (1.72)*		0.0047 (2.07)**	0.0047 (1.72)*		0.0047 (2.07)**	0.0047 (1.72)*
<i>Hold and Win 67</i>					-0.1359 (2.49)**	-0.0879 (0.91)		-0.0879 (0.91)	-0.0879 (0.91)		-0.0879 (0.91)	-0.0879 (0.91)		-0.0879 (0.91)	-0.0879 (0.91)
(<i>Hold and Win 67</i>) $\times$ (Cash flow)					0.3254 (3.83)***	0.2869 (3.45)***		0.2869 (3.45)***	0.2869 (3.45)		0.2869 (3.45)	0.2869 (3.45)		0.2869 (3.45)	0.2869 (3.45)
<i>Hold and Low 67</i>					-0.0289 (0.84)	-0.0622 (2.96)**		-0.0622 (2.96)**	-0.0622 (2.96)		-0.0622 (2.96)	-0.0622 (2.96)		-0.0622 (2.96)	-0.0622 (2.96)
(<i>Hold and Low 67</i>) $\times$ (Cash flow)					0.1417 (2.22)**	0.2366 (3.71)***		0.2366 (3.71)***	0.2366 (3.71)		0.2366 (3.71)	0.2366 (3.71)		0.2366 (3.71)	0.2366 (3.71)
Year-fixed effects	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Firm-fixed effects	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
(Year-fixed effects) $\times$ (Cash flow)	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
(Industry-fixed effects) $\times$ (Cash flow)	no	no	no	no	no	no	no	no	no	no	no	no	no	no	no

Table VI, first page: exercise behavior and investment

Table VI, continued

5.7 Table VII: Other overconfidence measures

Read:

- Table VII repeats the investment-cash flow regressions using Longholder and Net Buyer.
- Longholder increases investment-cash flow sensitivity in fixed-effects specifications.
- Net Buyer also increases investment-cash flow sensitivity, particularly with industry-cash flow interactions.
- The Net Buyer classification and investment outcomes use disjoint periods, reducing simultaneity concerns.

Economic message: The behavioral mechanism is robust across option-exercise and stock-purchase measures.

Table VII
Regression of Investment on Cash Flow and Longholder or Net Buyer

The dependent variable in the regressions is investment, defined as firm capital expenditures and normalized by capital at the beginning of the fiscal year. Cash flow is earnings before extraordinary items plus depreciation and is normalized by capital at the beginning of the fiscal year. Q is the market value of assets over the book value of assets at the beginning of the year. Stock ownership is the fraction of company stock owned by the CEO and his immediate family at the beginning of the year. Vested options are the CEO's holdings of options that are exercisable within 6 months of the beginning of the year, as a fraction of common shares outstanding. Vested options are multiplied by 10 so that the mean is comparable to stock ownership. Size is the natural logarithm of assets at the beginning of the year. Corporate governance is the number of outside directors who currently serve as CEOs of other companies. *Longholder* is a dummy variable equal to one if the CEO ever held an option until the last year prior to expiration. *Net Buyer* is a dummy variable equal to one if the CEO was a net buyer of stock more years than he was a net seller in his first five years in the sample. Columns 5-8 include only CEO years with at least 10 years in the sample and exclude their first five years. Industries are defined as the 12 Fama-French industry groups. Standard errors in columns 5, 4, 7, and 8 are robust to heteroskedasticity and arbitrary within-firm serial correlation.

	Longholder Regressions				Net Buyer Regressions				Industry-CF Interactions, Clustered by Firm			
	Fixed Effects, Controls (1)	Overconfidence with Fixed Effects, Controls (2)	Standard Errors Clustered by Firm (3)	Firm-CF Interactions, Clustered by Firm (4)	Fixed Effects, Controls (5)	Overconfidence with Fixed Effects, Controls (6)	Standard Errors Clustered by Firm (7)	Firm-CF Interactions, Clustered by Firm (8)	Industry-CF Interactions, Clustered by Firm (9)	Industry-CF Interactions, Clustered by Firm (10)	Industry-CF Interactions, Clustered by Firm (11)	Industry-CF Interactions, Clustered by Firm (12)
Cash flow	0.7249 (8.32)***	0.6056 (7.50)***	0.6056 (7.50)***	1.1063 (2.50)***	1.6092 (7.34)***	1.555 (2.50)***	1.555 (2.50)***	1.555 (2.50)***	1.555 (2.50)***	1.555 (2.50)***	1.555 (2.50)***	1.555 (2.50)***
Q	0.0814 (2.88)**	0.0851 (3.10)***	0.0851 (3.10)***	0.1009 (3.45)***	0.0752 (3.10)***	0.0770 (3.10)***	0.0770 (3.10)***	0.0770 (3.10)***	0.0770 (3.10)***	0.0770 (3.10)***	0.0770 (3.10)***	0.0770 (3.10)***
Stock ownership (%)												
Vested options												
Size												
Corporate governance												
$Q \times$ (Cash flow)												
(Stock ownership) $\times$ (Cash flow)												

Table VIII
Regression of Investment on Cash Flow and Overconfidence by Equity Dependence

The dependent variable in the regressions is investment, defined as firm capital expenditures and normalized by capital at the beginning of the fiscal year. Cash flow is earnings before extraordinary items plus depreciation and is normalized by capital at the beginning of the fiscal year. Q is the market value of assets over the book value of assets and is taken at the beginning of the year. Stock ownership is the fraction of company stock owned by the CEO and his immediate family at the beginning of the year. Vested options are the CEO's holdings of options that are exercisable within 6 months of the beginning of the year, as a fraction of common shares outstanding. Vested options are multiplied by 10 so that the mean is comparable to stock ownership. Size is the natural logarithm of assets at the beginning of the year. Corporate governance is the number of outside directors who currently serve as CEOs of other companies. *Longholder* is a dummy variable equal to one if the CEO ever held an option until the last year prior to expiration. Firms are classified according to quintiles of the Kaplan-Zingales index, where the highest quintile contains the most constrained subsample. All standard errors are robust to heteroskedasticity and arbitrary within-firm serial correlation.

	OLS with Fixed Effects				Least Constrained (5)			
	Most Constrained (1)	(2)	(3)	(4)	Least Constrained (5)	(6)	(7)	(8)
Cash flow	1.1538 (2.21)**	0.1763 (0.34)	0.8802 (2.08)**	0.5259 (2.50)**	0.8809 (2.50)**	0.8809 (2.50)**	0.8809 (2.50)**	0.8809 (2.50)**
Q			0.0068 (1.41)	0.0124 (4.48)***	0.0124 (4.48)***	0.0124 (4.48)***	0.0124 (4.48)***	0.0124 (4.48)***
Stock ownership (%)			0.5790 (4.03)	0.0066 (0.43)	0.0066 (0.43)	0.0066 (0.43)	0.0066 (0.43)	0.0066 (0.43)
Vested options			0.1101 (1.60)	0.15 (0.79)	0.15 (0.79)	0.15 (0.79)	0.15 (0.79)	0.15 (0.79)
Size			-0.0428 (0.50)	-0.0623 (1.43)	-0.0623 (1.43)	-0.0623 (1.43)	-0.0623 (1.43)	-0.0623 (1.43)
Corporate governance			0.0022 (0.21)	-0.0044 (0.64)	-0.0044 (0.64)	-0.0044 (0.64)	-0.0044 (0.64)	-0.0044 (0.64)
$Q \times$ (Cash flow)			0.0022 (2.12)**	0.0022 (0.30)	0.0022 (0.30)	0.0022 (0.30)	0.0022 (0.30)	0.0022 (0.30)

5.8 Table VIII: Equity dependence

Read:

- Table VIII splits firms by Kaplan-Zingales index quintiles.
- The Longholder-cash flow interaction is strongest and highly significant in the most constrained quintile.
- In the most constrained quintile, the Longholder-cash flow coefficient is about 0.499 and significant at the 1% level.
- The interaction is much smaller or insignificant in less constrained quintiles.

Economic message: The result supports Prediction 2: overconfidence matters most when the CEO must rely on external equity to finance investment.

5.9 Table IX: Personal characteristics and overconfidence

Read:

- Table IX adds personal CEO characteristics such as education, career background, birth cohort, and role concentration.
- Technical education or technical career backgrounds are associated with higher investment-cash flow sensitivity.
- Finance education or finance career backgrounds are associated with lower sensitivity.
- CEOs born in the 1930s and CEOs holding multiple positions show higher sensitivity.
- The overconfidence effect remains after adding these characteristics.

Economic message: CEO characteristics beyond standard firm controls matter for corporate investment. Overconfidence is not simply a stand-in for observable biography.

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Table IX
Regression of Investment on Personal Characteristics and Longholder

The dependent variable in the regressions is Investment, defined as firm capital expenditures and normalized by capital at the beginning of the year. Cash flow (CF) is earnings before extraordinary items plus depreciation and is normalized by capital at the beginning of the year. Q is the market value of assets over the book value of assets at the beginning of the year. Titles is a dummy variable equal to one for all CEO-years if the CEO is also president and chairman of the board. Tenure is the number of years the CEO has held that position. "Depression baby" is a dummy variable equal to one if the CEO was born in the 1930s. Finance Education is a dummy variable equal to one if the CEO had "financial education." Financial education includes undergraduate and graduate degrees in accounting, finance, business (incl. MBA), and economics. Technical Education is a dummy variable equal to one if the CEO had "technical education." Technical education includes undergraduate and graduate degrees in engineering, physics, operations research, chemistry, mathematics, biology, pharmacy, and other applied sciences. Longholder is a dummy variable equal to one if the CEO ever held an option until the last year prior to expiration. Industries are defined as the 12 Fama-French industry groups. All standard errors are robust to heteroskedasticity and arbitrary within-firm serial correlation.

	OLS with Fixed Effects				
	Titles (1)	Cohort 1930s (2)	Employment Background (3)	All Personal Characteristics (4)	All Personal Characteristics and Longholder (5)
Cash Flow	0.9414 (3.08)***	1.0181 (2.78)***	0.9946 (2.87)***	0.8087 (2.94)***	0.7106 (2.63)***
Q	0.0751 (1.04)	0.078 (1.08)	0.0868 (1.25)	0.0769 (1.05)	0.0819 (1.13)
Titles	-0.0239 (0.97)			-0.0208 (0.86)	-0.0199 (0.83)
(Titles) * (CF)	0.1342 (1.38)			0.1234 (1.30)	0.1222 (1.30)
Tenure	-0.0013 (0.88)	-0.0021 (1.20)		-0.0018 (1.18)	-0.0013 (0.93)
(Tenure) * (CF)	0.0032 (0.70)	0.0044 (0.89)		0.0043 (0.92)	0.0026 (0.60)
"Depression baby"		-0.0415 (1.33)		-0.0527 (1.88)*	-0.0432 (1.69)*
("Depression baby") * (CF)		0.1138 (1.32)		0.1427 (2.00)**	0.1255 (1.85)*
Finance Education			0.04 (1.62)	(2.26)**	0.0513 (2.32)**
(Finance Education) * (CF)			-0.123 (1.80)*	-0.1458 (2.34)**	-0.1482 (2.40)**
Technical Education			-0.0475 (1.78)*	-0.0496 (1.96)*	-0.0503 (2.07)**
(Technical Education) * (CF)			0.0894 (1.21)	0.1022 (1.44)	0.1117 (1.59)
Longholder					-0.0642 (1.65)
(Longholder) * (CF)					0.2196 (1.97)*
Observations	2201	2201	2201	2201	2201
Adjusted R^2	0.56	0.55	0.55	0.56	0.56

Constant included. Absolute value of t statistics in parentheses.

* Significant at 10%; ** significant at 5%; *** significant at 1%.

Controls for Corporate governance, Stock ownership, Vested options, Size and interactions of these variables and of Q with Cash Flow are included. Fixed effects for Year and Firm and the interactions of (Year) * (CF) and (Industry) * (CF) are also included.

6 Short summary

- The paper proposes CEO overconfidence as an explanation for investment-cash flow sensitivity.
- Overconfident CEOs overestimate project returns and believe equity is undervalued.
- They invest aggressively when internal funds are available and cut investment when external equity would be required.

- Overconfidence is measured from costly personal portfolio choices: late option exercise and net purchases of own-company stock.
- Late exercise is persistent and does not generate abnormal profits, weakening the private-information explanation.
- The main investment regressions show that overconfident CEOs have higher cash-flow sensitivity.
- The effect is strongest in equity-dependent firms.
- Results survive controls for Q, CEO stock ownership, vested options, firm size, governance, fixed effects, and other CEO traits.
- The paper shows that behavioral traits of top managers can have measurable corporate investment consequences.

7 Conclusion

7.1 Core conclusion

Malmendier and Tate show that CEO overconfidence helps explain investment-cash flow sensitivity. Overconfident CEOs do not merely respond to objective financing constraints; they perceive equity to be too costly because they overestimate the returns to their investment projects and the value of their firms.

7.2 Economic intuition

The intuition is that an overconfident CEO wants to invest more than the first-best level. If he has enough cash and debt capacity, he can pursue this desired investment level. If he must issue equity, he believes the market undervalues the firm and therefore avoids issuance. Cash flow relaxes this perceived constraint and raises investment.

7.3 Takeaway

The paper adds managerial beliefs to the investment literature. Investment distortions can arise even when managers try to maximize shareholder value and even when capital markets are efficient. Corporate governance may therefore need to constrain not only self-interested managers but also sincere, overconfident managers.